

RETAIL SALES DATA ANALYSIS REPORT

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Executive Summary:

This report summarizes the analysis of **1,000** retail transactions from January 2023 to December 2023 to identify revenue drivers and customer behavior patterns. The analysis states three critical insights:

1. **Balanced Revenue Distribution:**
Electronics (**34.41%**), Clothing (**34.12%**), and Beauty (**31.47%**) contribute nearly equally to total revenue of **\$456,000**, indicating a diversified portfolio with low category dependency risk.
2. **Female Customer Dominance:**
Female customers generate **51.06% (\$232,840)** of total revenue, making them the primary target for marketing initiatives. The Mature (46-55) and Adult (26-35) age groups collectively contribute **43.7%** of sales.
3. **Seasonal Opportunities:**
May is the peak month (**\$53,150**), while September shows the lowest performance (**\$23,620**). This presents a clear opportunity for targeted promotions to smooth seasonal fluctuations.

Project Introduction:

1. **Project Title:**
Retail Sales Data Analysis
2. **Objective:**
Analyze 1,000 retail transactions to identify revenue drivers, customer behavior patterns, and product category performance for strategic business recommendations.
3. **Dataset Overview:**
 - Source: Kaggle
 - Scope: 1,000 unique transactions
 - Key Attributes: Transaction ID, Date, Customer ID, Gender, Age, Product Category, Quantity, Price per Unit, Total Amount

Tools Used:

- Google Sheets: Data cleaning, pivot tables, KPI calculations, and dashboard creation
- Functions: SUM, AVERAGE, COUNTIF, Pivot Tables, Charts (Bar, Line, Pie)

Methodology:

1. **Data Inspection**
 - a. Total transactions: 1,000

- b. Date Range: 01/01/2023 - 31/12/2023
- c. Gender: Male & Female
- d. Age: 18 years - 64 years
 - 18 - 25 – Young Adult
 - 26 - 35 – Adult
 - 36 - 45 – Mid Age
 - 46 - 55 – Mature
 - 56 - 64 – Senior
- e. Product Category:
 - Beauty
 - Clothing
 - Electronics

2. Data Validation

Checked and all calculations are consistent

Total Amount = Quantity × Price per Unit

3. Data Cleaning Strategy

- a. No duplicates found
- b. No missing values
- c. Standardized date format

4. KPI Calculations

- a. KPI 1: Total Transactions = 1,000
- b. KPI 2: Total Revenue = \$ 456,000
- c. KPI 3: Total Units Sold = 2,514
- d. KPI 4: Average Revenue Per Unit = \$ 181.38
- e. KPI 5: Average Transaction Value = \$456

5. Exploratory Analysis

- a. Product Category Performance Analysis

Product	Revenue	Contribution
Beauty	\$143,515	31.47%
Clothing	\$155,580	34.12%
Electronics	\$156,905	34.41%

b. Customer Demographic Analysis

i. By Gender

Gender	Revenue	Contribution
Male	\$223,160	48.94%
Female	\$232,840	51.06%

ii. By Age

Age Group	Age Group Label	Revenue	Contribution
18 - 25	Young Adult	\$84,550	18.54%
26 - 35	Adult	\$98,480	21.59%
36 - 45	Mid-Age	\$91,870	20.14%
46 - 55	Mature	\$100,690	22.08%
56 - 64	Senior	\$80,410	17.63%

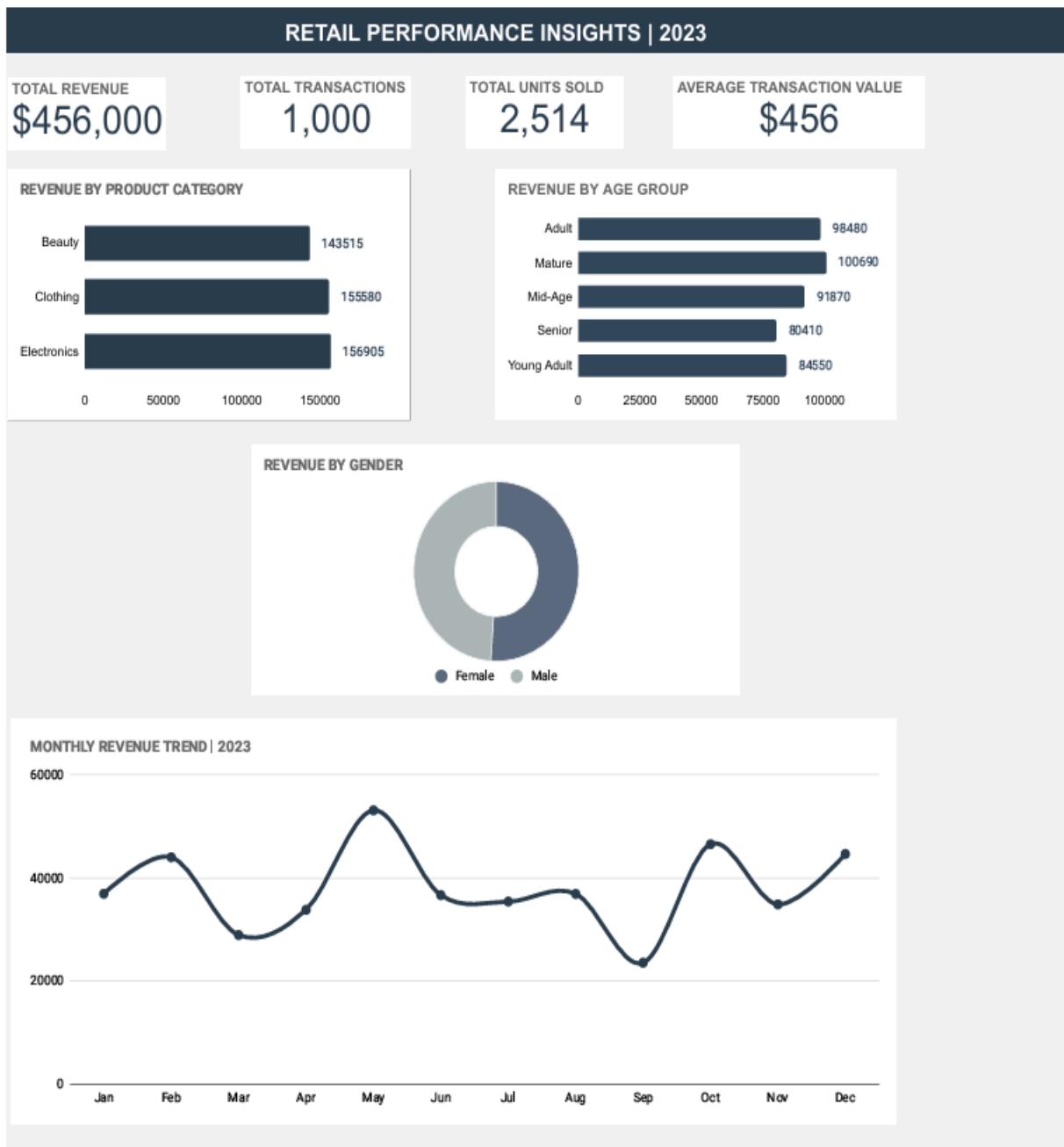
c. Monthly Trend Analysis

Month	Revenue	Contribution
January	\$36,980	8.11%
February	\$44,060	9.66%
March	\$28,990	6.36%
April	\$33,870	7.43%
May	\$53,150	11.66%
June	\$36,715	8.05%
July	\$35,465	7.78%
August	\$36,960	8.11%
September	\$23,620	5.18%
October	\$46,580	10.21%
November	\$34,920	7.66%
December	\$44,690	9.80%

6. Dashboard Development

Created an interactive dashboard in Google Sheets featuring -

- KPI Summary Cards
- Category Performance Bar Chart
- Age Group Revenue Bar Chart
- Gender Distribution Pie Chart
- Monthly Sales Trend Line Chart



Key Insights:

1. Revenue is Evenly Distributed Across Categories (Low Risk)
No single category dominates revenue. The business has a diversified product portfolio with low dependency risk on any one category.
2. Female Customers Drive the Majority of Revenue
Female customers generate **51.06%** of total revenue (**\$232,840**). Marketing and product strategies should prioritize this segment.
3. Mature and Adult Age Groups Are the Highest Spenders
The Mature (46-55) and Adult (26-35) segments together contribute **43.7%** of all revenue - nearly half. These are the high-value customer segments to target.
4. Sales Are Highly Seasonal — May Peaks, September Dips
Sales fluctuate significantly throughout the year. September represents a major opportunity for promotional intervention, while May and (Oct-Dec) are natural high-performance periods to maximize.
5. Average Transaction Value Indicates Healthy Ticket Size
The average customer spends **\$456** per transaction. This benchmark can be used to measure future campaign performance.

Growth Opportunities & Recommendations:

Based on the key insights from this analysis, the following recommendations are proposed:

1. Launch September Promotional Campaign.
2. Target Female Customers with Dedicated Campaigns.
3. Prioritize Marketing to Mature & Adult Age Groups.
4. Maximize October-December Performance.
5. Maintain Product Category Diversification.
6. Conduct Deeper Customer Analytics.